

AI-Driven Multi-Domain Threat Detection and Response

Nguyen Ngoc Bach, Tran Van Khoi, Pham Quang Huy, and Nguyen Duc Binh

Group 5

Member A – NLP Phishing & Web Attack

Member B – Continuous Authentication

Member C – Multimodal Deepfake Detection

Member D – Threat Correlation & Autonomous Security Agent

Lecturer: Le Minh Duc

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EXECUTIVE SUMMARY

This group report proposes a unified, AI-driven security system that performs *multi-domain* threat detection and *policy-bound* response by integrating four specialized modules: (A) NLP-based phishing and web-attack detection, (B) behavior-based continuous authentication, (C) multimodal deepfake detection (audio-visual), and (D) a centralized threat correlation engine with an autonomous security agent. The motivating operational reality is that attacks increasingly span multiple data modalities (e-mail, web traffic, identity behavior, and manipulated media), while security operations still struggle with alert overload, fragmented telemetry, and slow triage. Industry reporting continues to show large-scale incident volumes and persistent prevalence of social engineering and identity-centric intrusion paths, reinforcing the urgency of cross-signal correlation rather than isolated single-task classifiers [1]–[3].

Template compliance and scope. The provided course template defines required content for (i) *Section 1: Introduction and Background* and (ii) *Section 6: References* (and associated guidance). This submission intentionally includes *only* those template sections, plus standard IEEE front matter (title, abstract) and this executive summary to make the document submission-ready. All other template chapters (e.g., literature review chapter, design/implementation chapter, results/discussion chapter) are treated as out-of-scope by explicit instructor constraint.

Deliverable structure. Section 1 is expanded into four subsections aligned with the project title: problem description, research objectives, scope, and contribution structure. Within Section 1.4, each member contributes an embedded, approximately five-page module plan that includes: a 10-source literature synthesis, datasets and primary links, a proposed methodology with hyperparameter ranges and evaluation protocols, risk-event outputs, a five-week Gantt plan, and expected contributions. Section 6 includes conclusions, limitations, future work, a responsibility summary, and a consolidated IEEE reference list (40+ sources) with direct URLs/DOIs.

System integration stance. The unified system is designed to be implementable using a streaming and microservice stack (Kafka for event streaming, FastAPI for inference services, Docker for deployment isolation, and MLflow for experiment

tracking and model registry governance) [11]–[15].

Ethics and security. Because the system handles potentially sensitive data (e.g., email corpora, behavioral biometrics, face videos) and embeds automation, the report explicitly addresses data governance and adversarial AI risks using authoritative guidance and threat knowledge bases (NIST incident handling and continuous monitoring guidance, NIST AI RMF, MITRE ATT&CK/ATLAS) [4]–[6], [8], [10].

Abstract—This report proposes an AI-driven, multi-domain security system for threat detection and response, addressing limitations of siloed detectors and rule-heavy correlation in modern security operations. The system integrates four domain-focused modules: NLP-based phishing and web-attack detection, behavior-based continuous authentication, multimodal audio-visual deepfake detection, and a centralized threat correlation engine with a policy-bound autonomous response agent. Each module emits standardized risk events that are fused into incident hypotheses aligned with threat knowledge bases and organizational policy constraints. The proposed architecture is designed for near-real-time operation using event streaming and containerized microservices, with experiment tracking and model registry governance for reproducibility and safe deployment. Evaluation is planned using public benchmark datasets across domains, employing task-appropriate metrics (F1/ROC-AUC for text/web, FAR/FRR/EER for authentication, cross-dataset robustness for deepfakes) and system-level measures (latency, false-positive reduction, and response safety). The principal contribution is a coherent, submission-ready research plan demonstrating how heterogeneous AI detectors can be integrated into a unified correlation-and-response pipeline with explicit governance and measurable evaluation protocols.

Index Terms—Multi-domain security, phishing detection, web attack detection, continuous authentication, behavioral biometrics, deepfake detection, multimodal learning, threat correlation, autonomous response, MLOps.

I. INTRODUCTION AND BACKGROUND

A. Problem Description

1) *Real-world problem and context:* Security operations must make timely, high-stakes decisions from heterogeneous and high-velocity signals: e-mail content and headers, URLs and web requests, identity and behavioral telemetry, and audio-visual media shared via collaboration platforms. Attackers increasingly orchestrate multi-stage chains that combine social

engineering, credential abuse, web exploitation, and impersonation. Large-scale incident reporting continues to document high incident volumes and persistent breach drivers, while phishing trend reporting highlights sustained and evolving phishing campaign activity [1]–[3].

At the same time, defenders face “alert fatigue” due to fragmented systems: detectors are often built per-domain and per-vendor with limited interoperability and weak cross-domain correlation. This leads to (i) duplicated alerts, (ii) inconsistent severity scores, (iii) delayed analyst confirmation, and (iv) slow or risky response automation. Moreover, multi-modal threats (e.g., deepfake-enabled impersonation supporting an account takeover) cannot be reliably assessed by any single-domain model, motivating a unified, correlation-centered design.

2) *Research gap*: Academic work has achieved strong task-specific models in phishing detection, URL analysis, HTTP attack detection, behavioral biometric authentication, and deepfake detection. However, three gaps limit operational usefulness:

- 1) **Cross-domain correlation gap**: decision-making remains under-specified when multiple weak signals should jointly raise confidence (e.g., “phish click” + anomalous behavior + suspected deepfake).
- 2) **Real-time system gap**: many papers evaluate offline classification without latency-aware streaming integration or governance requirements.
- 3) **Response governance gap**: automation is either absent or under-constrained; incident response guidance emphasizes systematic handling and safety but does not prescribe how to operationalize AI-driven recommendations under policy constraints [4], [5].

3) *Why this is urgent*: Threat knowledge bases show that real-world adversary behavior spans multiple tactics and techniques requiring multiple evidence sources for confident attribution and prioritization [8]. Additionally, AI introduces new attacker capabilities (e.g., manipulation and scalable social engineering) and new defender risks (e.g., model extraction and adversarial inputs), requiring structured AI risk governance [6], [10].

4) *System overview and architecture (brief, unified)*: Although the template restricts the report to Section 1 and Section 6, a brief architecture overview is necessary to make Section 1 meaningful and coherent, and to describe how the four member modules form a unified system.

a) *Mermaid architecture flowchart (render externally)*:

To satisfy the “Mermaid diagram” requirement in a submission-ready manner, the code below should be rendered using a Mermaid-compatible renderer (e.g., <https://mermaid.live>) and exported as a PDF/SVG/PNG, then included as a figure in the Overleaf project.

b) *Architecture image placeholder (include in Overleaf)*:

c) *Integration stack (system-level)*: The integration plan uses an event streaming backbone and containerized microservices to support near-real-time ingestion and scalable inference. Kafka provides topics/partitions/consumer groups for durable event pipelines [11]. Model lifecycle management

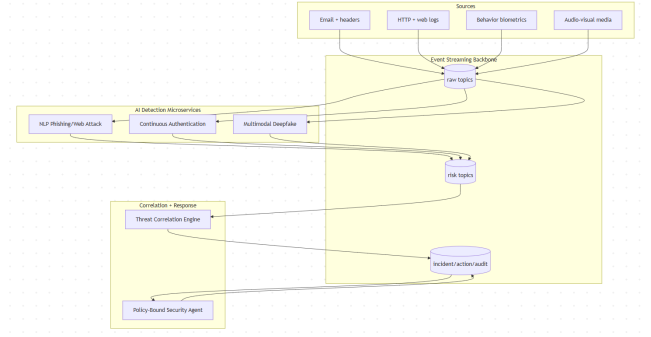


Fig. 1. System architecture and event flow.

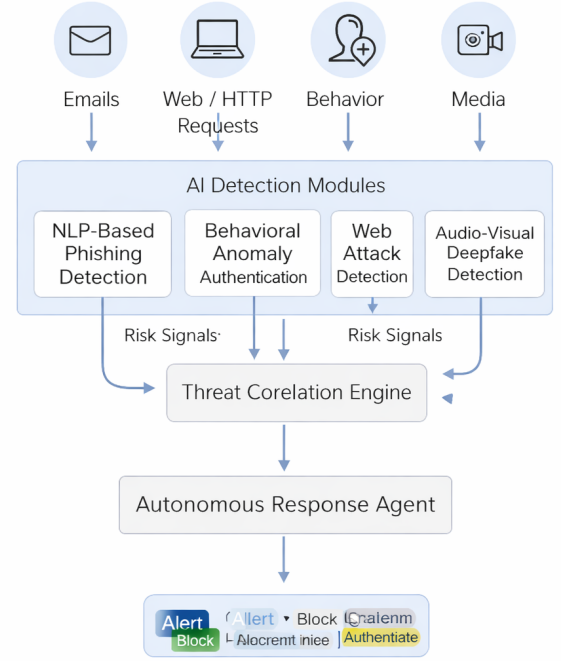


Fig. 2. System overview (placeholder). Replace with an exported image rendered from Listing ??.

is handled via MLflow tracking and model registry to enforce lineage, reproducible training, and staged promotion [14], [15]. FastAPI is used for inference APIs with OpenAPI-based interfaces and typed validation [12]. Containers provide consistent packaging and deployment isolation [13].

B. Research Objectives

1) *SMART objectives*: The project’s objectives are formulated to be specific, measurable, achievable, relevant, and time-bound (time horizon: five weeks per member module plan; system integration is specified at the interface level in this report).

O1 (Unified risk-event schema and interfaces). Define a shared schema for risk events emitted by all detection modules, including required identification and audit fields (model version, feature lineage, timestamps, and artifact pointers).

O2 (Module-level research plans). Develop four module research plans with reproducible dataset sourcing, explicit

preprocessing, candidate models, hyperparameter ranges, and evaluation protocols. Each module must (i) summarize 10 high-quality sources, and (ii) propose a five-week work plan.

O3 (Correlation and response plan). Specify a centralized correlation engine and policy-bound response agent design that consumes standardized risk events and produces incident hypotheses aligned to ATT&CK terminology and incident handling guidance [4], [8].

O4 (Governance, safety, and ethics). Document governance controls for AI risk, data handling, and automation safety, including mitigations for adversarial ML threats such as model extraction attacks [6], [16].

2) *Research questions (as required by template):* The template requires explicit research questions for: methods (literature), proposed method/technology, and evaluation plan. Accordingly, our research questions are:

RQ1 (Methods). What are the leading methods in each threat domain (phishing/web attacks, continuous authentication, multimodal deepfakes, alert correlation/response), and what are their empirical strengths and limitations in real-world deployment?

RQ2 (Proposed system). What integrated architecture and interfaces enable heterogeneous AI modules to jointly support incident-level reasoning, while providing reproducibility and governance?

RQ3 (Evaluation). What module-level and system-level evaluation protocols best demonstrate operational value beyond single-domain baselines (accuracy, calibration, robustness, latency, and response safety)?

C. Scope of the Project

1) *Included scope:* This report focuses on *planning and specification* consistent with the course template constraints. The included scope is:

- 1) Multi-domain detection module plans: NLP phishing/web, continuous authentication, multimodal deepfakes, correlation/response.
- 2) Public dataset sourcing with primary links (e.g., Enron, SpamAssassin, CIC-IDS2017, CSIC 2010, FaceForensics++, DFDC, FakeAVCeleb, ASVspoof, keystroke benchmarks).
- 3) Unified risk-event schema and Kafka topic plan for integration.
- 4) An MLOps and governance plan (experiment tracking, model registry, audit logging).
- 5) Security and ethical considerations (PII minimization, bias and distribution shift, adversarial ML).

2) *Excluded scope:* Because the instructor restricted the submission to Section 1 and Section 6, the following are excluded in this report:

- 1) Full system design/implementation details and code-level integration (out-of-scope template chapters).
- 2) Full experimental results and ablation outcomes (out-of-scope template chapters).
- 3) Production deployment integration with proprietary SOC platforms (requires organizational infrastructure).

TABLE I
COMPUTE/RESOURCE OPTIONS (BUDGET UNSPECIFIED).

Option	Practical planning assumption
Minimal	Single workstation: 1 GPU (16–24GB VRAM), 16–32 CPU cores, 64–128GB RAM; suitable for fine-tuning base transformers, moderate CV backbones with reduced resolution/frames, limited hyperparameter sweeps.
Moderate	Lab server: 2–4 GPUs (24–48GB VRAM); supports larger batch sizes, multi-run ablations, broader robustness evaluation.
High	Shared cluster: 4–8 high-end GPUs; enables systematic cross-dataset evaluation, extensive ablations, and heavier multimodal architectures.

TABLE II
RESPONSIBILITY MATRIX MAPPING MEMBERS TO DELIVERABLES.

Member	Primary responsibility (embedded module in Section 1.4)
A	NLP-based Phishing & Web Attack Detection (risk.nlp); focuses on email/HTTP semantics, phishing webpages/URLs, and web attack strings.
B	Behavior-based Continuous Authentication (risk.auth); focuses on session-level identity risk from keystroke/mouse/touch/sensor sequences.
C	Multimodal Deepfake Detection (risk.deepfake); focuses on audio-visual manipulation detection and cross-modal consistency.
D	Threat Correlation Engine & Autonomous Security Agent (risk.corr / incidents/actions); fuses risk events into incidents and policy-bound actions.

3) *Data governance, security, and ethical considerations (system-level):*

a) *Sensitive data handling.:* Even when using public datasets, e-mail collections and biometric datasets can contain sensitive content. Therefore, the system plan includes access control, hashing/pseudonymization of identifiers, and limiting raw artifact retention. Dataset use shall comply with dataset terms and institutional policies.

b) *AI risk governance and adversarial ML.:* NIST AI RMF provides a lifecycle-oriented framing for AI risk management, emphasizing governance and continuous evaluation [6]. For externally exposed inference services, model extraction attacks demonstrate that naive output interfaces can leak model behavior even when confidence scores are restricted [16]. Adversarial example literature further shows that small perturbations can cause misclassification, motivating robustness testing and monitoring [17].

c) *Automation safety in incident response.:* Incident response guidance emphasizes systematic response stages and risk-aware decision-making across preparation, detection/analysis, containment, eradication/recovery, and improvement [4]. We therefore constrain the autonomous response agent to be policy-bound with approval gates for high-impact actions.

4) *Compute/resource assumptions (explicitly unspecified):* The compute budget is **unspecified**. Therefore, we provide three reasonable options for planning and reproducibility.

D. Contribution Structure

1) *Responsibility matrix (system-level):*

2) *Unified risk-event schema (system contract):* To enable correlation, all modules emit a standardized *risk event*.

TABLE III
UNIFIED RISK-EVENT SCHEMA (MINIMUM REQUIRED FIELDS).

Field	Description
event_id	Unique event identifier (UUID).
ts_utc	Event timestamp (UTC).
source_domain	{email, http, auth, media}.
subject_id	Pseudonymous user/account/device identifier.
asset_id	Optional asset identifier (service/app/endpoint).
artifact_type	{email, url, http_request, session_window, media}.
artifact_ref	Pointer to stored artifact (hash+URI), not raw content.
model_id, model_ver	Model name and version (MLflow registry ID).
risk_score	Calibrated score in [0,1].
risk_label	Predicted label (domain-specific taxonomy).
confidence	Optional confidence or calibrated probability.
explain_ref	Pointer to explanation artifact (e.g., token highlights).
trace_id	Correlation tracing ID for distributed systems.
schema_ver	Schema version for compatibility.

TABLE IV
INTEGRATION STACK AND RATIONALE.

Component	Role in the unified system
Kafka	Durable event streaming topics for raw events and risk events; enables consumer-group scaling and backpressure handling [11].
FastAPI	Inference microservice APIs with typed validation; supports OpenAPI docs for consistent contracts [12].
Docker	Reproducible runtime packaging and isolation per service; simplifies deployment and dependency control [13].
MLflow	Experiment tracking, artifact logging, and model registry for lineage, versioning, and staged deployment governance [14], [15].

Table III defines required fields (minimum viable schema). Module-specific fields extend this schema but must not break backwards compatibility; enforcement is performed via schema versioning.

3) *Integration stack plan (Kafka, FastAPI, Docker, MLflow)*: Table IV summarizes the integration stack and its role in governance and reproducibility.

4) *Member A Module (5-page contribution): NLP-based Phishing & Web Attack Detection*:

a) *Module Title.*: **NLP-based Phishing and Web Attack Detection from Email + HTTP Request Semantics.**

b) *Problem Definition.*: Phishing remains a primary initial-access vector and increasingly blends high-quality language with realistic branding and multi-step interaction. In parallel, web applications remain exposed to attacks embedded in HTTP requests (e.g., SQL injection and XSS), where rule-only detection can be brittle and produce high false positives. The objective is to design an NLP-centric detector that (i) classifies phishing emails (and optionally phishing webpages/URLs), and (ii) detects web attacks from HTTP request strings, generating calibrated risk events for downstream correlation.

c) *Input Data (datasets + primary links).*: Member A uses public datasets and official sources:

- 1) **Enron Email Dataset (benign corpora)**: CMU CALO release page <https://www.cs.cmu.edu/~enron/> [28].
- 2) **SpamAssassin Public Corpus (spam/ham baseline)**: Apache SpamAssassin corpus description

<https://spamassassin.apache.org/old/publiccorpus/readme.html> [29].

- 3) **CIC-IDS2017 (web attack scenarios, labeled flows)**: CIC dataset page <https://www.unb.ca/cic/datasets/ids-2017.html> [30].
- 4) **HTTP CSIC 2010 (HTTP request anomalies)**: Impact-CyberTrust dataset page https://www.impactcybertrust.org/dataset_view?idDataset=940 [31].

Justification. Enron/SpamAssassin support email text baselines and realistic lexical variability; CIC-IDS2017 provides intrusion contexts including web attack labels and enables cross-domain correlation with flow-level signals; CSIC 2010 provides request-level payload-like patterns for HTTP anomaly classification.

d) *Methodology (architectures, preprocessing, hyperparameter ranges, splits).*: **Model families.**

- 1) **Transformer encoder for email/phishing text**: Fine-tune a pretrained Transformer (BERT/Roberta) with a classification head for binary or multiclass labels [18], [19].
- 2) **Character-aware CNN/Transformer for HTTP requests**: Model HTTP request strings (method/path/query/headers subset) as a token+character sequence; include character-level convolution or subword tokenization to handle obfuscation and encoding variants.

Preprocessing. Email: MIME decoding, HTML stripping (retain text), URL extraction, normalization (lowercase where appropriate), header feature extraction (From/Reply-To mismatch flags, domain similarity). HTTP: URL decoding, HTML entity decode, canonicalization (normalize parameter order), tokenize by separators, optionally mask numeric IDs, and maintain raw substrings for character-CNN.

Training/validation split. Email: stratified split (e.g., 80/10/10) with a *campaign-disjoint* guard when possible (group by sender domain or near-duplicate template clusters to avoid leakage). HTTP: time- or source-disjoint where possible; for CSIC 2010, request-level stratified splits and robustness tests under encoding variants.

Hyperparameter ranges (planned). Transformer fine-tuning: learning rate $[1 \times 10^{-5}, 5 \times 10^{-5}]$, batch size [8, 64], epochs [2, 10], weight decay [0, 0.1], max sequence length [128, 512]. Imbalance handling: class weights or focal loss ($\gamma \in [1, 3]$). Calibration: temperature scaling on validation.

e) *Output (risk schema fields).*: Emit `risk.nlp` events extending Table III with:

- `extracted_iocs`: list of URLs/domains/keywords (hashed or normalized)
- `text_span_highlights`: pointer to token highlight artifact (`explain_ref`)
- `request_features`: optional compressed summary (e.g., length, entropy, special-char ratio)

f) *Integration Role (feed correlation engine).*: Member A's risk events provide *initial-access and web exploitation* evidence. The correlation engine links extracted indicators (URLs/domains) with subsequent HTTP events and identity anomalies. Example: a suspicious email containing a URL fol-

TABLE V
MEMBER A: METRICS PLAN (MODULE-LEVEL).

Category	Metrics
Effectiveness	Precision, Recall, F1 (macro/weighted), ROC-AUC
Calibration	ECE, reliability plots (offline)
Robustness	Performance under encoding/obfuscation variants
Runtime	p50/p95 inference latency; throughput in msgs/sec



Fig. 3. System architecture and event flow.

lowed by an HTTP request to the same domain with suspicious parameters increases incident confidence and severity.

g) *Evaluation metrics and protocols.*: Email: precision/recall/F1 (macro + weighted), ROC-AUC, and expected calibration error (ECE). HTTP: per-attack-type F1 (SQLi/XSS/other), confusion matrices, false-positive rate at fixed recall. Robustness: test-time perturbations (URL encoding variants, obfuscated payload tokens). Latency: p50/p95 inference time under batch-1 microservice constraints.

h) *Five-week timeline (Mermaid source).*:

i) *Expected contributions.*: (1) A unified NLP detector producing calibrated `risk.nlp` events across email and HTTP strings; (2) quantified robustness and calibration results; (3) correlation-friendly IoC extraction supporting multi-domain incident reasoning.

j) *Literature review (10 sources; 1–2 sentence summary each).*:

- 1) **BERT** [18]: Establishes bidirectional Transformer pre-training and fine-tuning as a strong general approach for text classification, motivating phishing email classification using contextual embeddings.
- 2) **RoBERTa** [19]: Demonstrates that pretraining and hyperparameter choices significantly affect performance; informs careful tuning and reproducible baselines.
- 3) **URLNet** [20]: Proposes end-to-end URL representation learning from character and word CNNs; highlights benefits over manual lexical features for detecting unseen malicious URLs.
- 4) **PhishBench 2.0** [21]: Provides a benchmarking framework enabling consistent evaluation of phishing systems; motivates leakage-aware and comparable evaluation practices.
- 5) **BERT-based phishing email detection** [22]: Reports fine-tuning compact and standard Transformer variants for phishing emails; reinforces that dataset composition and tuning strongly influence outcomes.
- 6) **Phishpedia (USENIX 2021)** [23]: A hybrid deep learning system combining logo detection and Siamese brand matching; shows that visual/brand semantics can reduce false positives for phishing webpages.
- 7) **PhishIntention (USENIX 2022)** [24]: Infers phishing intention via webpage appearance and hybrid analysis; reports large false-alert reductions in field study scenarios relative to baselines.

- 8) **DynaPhish (USENIX 2023)** [25]: Extends reference-based detection by dynamically expanding references with legitimacy validation and counterfactual explanations; motivates runtime adaptability and explainability.
- 9) **Benchmark dataset scheme for phishing** [26]: Proposes guidelines for reproducible phishing website datasets and shows feature/runtime trade-offs; informs dataset design and measurement practices.
- 10) **Improving WAFs via anomaly detection** [27]: Shows how one-class learning and n-gram analysis can complement rule-based WAFs (ModSecurity/OWASP CRS) and reduce false positives in practice-oriented settings.

5) *Member B Module (5-page contribution): Behavior-based Continuous Authentication:*

a) *Module Title.*: **Behavior-based Continuous Authentication via Keystroke/Mouse/Touch/Sensor Sequences.**

b) *Problem Definition.*: One-time authentication is insufficient for long-lived sessions because credentials can be compromised, sessions hijacked, and insiders misuse access. Continuous authentication (CA) attempts to verify identity throughout a session using behavioral biometrics. The challenge is that behavioral signals drift (fatigue, device change, context) and evaluation must reflect verification performance (FAR/FRR/EER) rather than only classification accuracy [32], [33].

c) *Input Data (datasets + primary source links).*:

Member B’s dataset plan emphasizes public benchmarks and repeatable protocols:

- 1) **CMU Keystroke Dynamics Benchmark (DSL)**: dataset and benchmark references <https://www.cs.cmu.edu/~keystroke/> [42].
- 2) **Balabit Mouse Dynamics Challenge**: official repository <https://github.com/balabit/Mouse-Dynamics-Challenge> [43].
- 3) **Optional smartphone sensor CA datasets (if available under time)**: use published sensor-based CA studies’ release artifacts when accessible; otherwise treat as out-of-scope for implementation and use as literature guidance.
- d) *Methodology (architectures, preprocessing, hyperparameter ranges, splits).*: **Operational modes.**

- 1) **Verification mode**: confirm that the current session window matches an enrolled user profile.
- 2) **Takeover detection mode**: detect anomalous deviation from baseline behavior to trigger step-up authentication.

Candidate architectures.

- 1) **Sequence encoders (LSTM/Transformer)**: represent event sequences (keystroke dwell/flight; mouse velocity/acceleration) and output a session embedding; classification can be Siamese or softmax verification [35], [36].
- 2) **Lightweight convolutional-transformer (mobile-feasible)**: adopt deployment-aware designs for sensor sequences [37].
- 3) **Siamese CNN for sensor signatures**: prior work shows Siamese learning for motion/sensor authentication [38].

Preprocessing. Windowing (5–30s rolling windows), per-user normalization, outlier filtering, and drift indicators (feature distribution shift). For keystrokes: compute dwell and

TABLE VI
MEMBER B: METRICS PLAN (MODULE-LEVEL).

Category	Metrics
Verification	FAR, FRR, EER
Takeover detection	Time-to-detect, detection delay distribution
Robustness	Drift sensitivity; performance over time windows
Runtime	p50/p95 inference time per window



Fig. 4. System architecture and event flow.

flight times; for mouse: segment into move/click/drag actions and compute speed/angle features.

Training/validation split. *Strict separation of enrollment and test windows* to prevent leakage. Evaluate both user-dependent and user-independent scenarios where applicable. For large-scale free-text scenarios, follow identity disjoint splits as in TypeNet [35].

Hyperparameter ranges (planned). Embedding dimension [64, 512], sequence length [50, 500] events, learning rate $[10^{-4}, 10^{-3}]$ (non-pretrained), batch size [32, 512], dropout [0.1, 0.5]. Thresholds selected on dev set by minimizing EER; report confidence intervals across seeds.

e) *Output (risk schema fields).*: Emit `risk.auth` events extending Table III with:

- `window_seconds`, `events_count`
- `auth_score` (0–1), decision (allow/step-up/revoke)
- `drift_score` (distribution shift indicator)
- `eer_threshold_id` (configuration pointer)

f) *Integration Role (feed correlation engine).*: The authentication module provides identity-centric risk signals to corroborate or refute other evidence. Example: suspicious email link click (Member A) plus anomalous CA score suggests account takeover and justifies immediate step-up authentication rather than only alerting.

g) *Evaluation metrics and protocols.*: Primary: FAR, FRR, and EER (verification). Secondary: takeover detection delay (time from hijack to threshold crossing), stability under drift (performance over time), and fairness checks when demographic metadata exist (often limited in public datasets). Follow survey guidance on CA evaluation pitfalls [33], [34].

h) *Five-week timeline (Mermaid Gantt).*:

i) *Expected contributions.*: (1) A continuous authentication plan with verification-grade metrics (FAR/FRR/EER) and takeover delay evaluation; (2) a standardized `risk.auth` event contract; (3) explicit drift-aware evaluation and threshold governance.

j) *Literature review (10 sources; 1–2 sentence summary each).*:

- 1) **Keystroke benchmark (DSN 2009)** [32]: Establishes a comparative benchmark for keystroke anomaly detection and shows that inconsistent evaluation invalidates prior comparisons; motivates protocol rigor.

- 2) **Behavioral biometrics CA survey (IEEE Access)** [33]: Surveys AI-based CA methods and defines the standard metrics and deployment challenges (drift, heterogeneity).
- 3) **Continuous authentication survey (IEEE IoT J.)** [34]: Surveys 140+ approaches and highlights modality trade-offs and multimodal CA; informs feasibility and risk analysis.
- 4) **TypeNet (100K users)** [35]: Demonstrates large-scale free-text keystroke biometrics and offers insights into scaling and evaluation across populations.
- 5) **Siamese keystroke verification (Sensors 2023)** [36]: Uses Siamese networks for keystroke verification and supports n-shot verification framing for recent-session matching.
- 6) **AuthConFormer (Computers & Security 2023)** [37]: Proposes a convolutional-transformer for continuous smartphone authentication and emphasizes deployment feasibility.
- 7) **Siamese CNN mobile CA (MobiSys workshop 2018)** [38]: Shows Siamese CNN learning of motion signatures, motivating similarity-based CA.
- 8) **DeepAuthen (Sensors 2021)** [39]: Introduces a deep learning CA architecture using smartphone sensors; useful for modeling drift and sensor fusion.
- 9) **Touch+sensor deep learning CA (Procedia 2019)** [40]: Studies combining touchscreen and sensor features using deep learning; supports multimodal feature design decisions.
- 10) **DL Siamese framework (Computers & Security 2025)** [41]: Proposes a keystroke-based authentication framework handling password variability; informs model design for variable-length sequences.

6) *Member C Module (5-page contribution): Multimodal Deepfake Detection (Audio-Visual):*

a) *Module Title.*: **Multimodal Deepfake Detection via Audio-Visual Consistency and Articulatory Representation Learning.**

b) *Problem Definition.*: Deepfake and manipulated media can enable impersonation, fraud, and disinformation. Visual-only detectors often overfit to artifacts and degrade under compression or unseen manipulations. Audio-visual deepfakes introduce coordinated manipulation across modalities; detectors must assess cross-modal consistency (lip sync, articulatory correlation) and generalize across datasets. Benchmark work emphasizes the need for standardized evaluation and robustness under distortions [44], [45], [48].

c) *Input Data (datasets + primary source links).*:

- 1) **FaceForensics++**: benchmark paper and PDF https://openaccess.thecvf.com/content_ICCV_2019/papers/Rossler_FaceForensics_Learning_to_Detect_Manipulated_Facial_Images_ICCV_2019_paper.pdf [44].
- 2) **DFDC**: dataset paper <https://arxiv.org/abs/2006.07397> [45].
- 3) **FakeAVCeleb**: dataset paper PDF <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/file/d9d4f495e875a2e075a1a4a6e1b9770f-Paper-round2.pdf> [46].

TABLE VII
MEMBER C: METRICS PLAN (MODULE-LEVEL).

Category	Metrics
Effectiveness	ROC-AUC, F1, balanced accuracy
Generalization	Cross-dataset performance drop; cross-manipulation tests
Robustness	Compression/noise sensitivity; ablations
Runtime	p50/p95 inference latency per clip/window

4) **ASVspoof 2019**: challenge paper (arXiv) <https://arxiv.org/abs/1911.01601> [47].

d) *Methodology (architectures, preprocessing, hyperparameter ranges, splits)*.: **Two-branch multimodal model**.

- 1) **Visual branch**: detect facial manipulation and temporal inconsistencies using a CNN backbone (e.g., Xception) and optional temporal head [44], [49], [50].
- 2) **Audio branch**: detect synthesized/converted/replayed speech characteristics using log-mel encoders and anti-spoofing baselines [47].
- 3) **Fusion**: (baseline) late fusion of calibrated scores; (advanced) cross-modal attention or articulatory representation learning to explore audio-visual correspondences [52].

Preprocessing. Video: face detection/tracking; crop and align; sample $T \in \{16, 32, 64\}$ frames per clip; compressions for robustness testing. Audio: resample (e.g., 16kHz), compute log-mel spectrograms (e.g., 80 bins), window length 2–6s aligned to video clip.

Training/validation split and generalization protocol. Use within-dataset split (e.g., 80/10/10) plus cross-dataset evaluation (train on FaceForensics++, test on DFDC/-FakeAVCeleb where feasible). Use cross-manipulation evaluation when dataset labels permit. For robustness, evaluate under compression/noise and report generalization gap.

Hyperparameter ranges (planned). Learning rate $[10^{-5}, 10^{-4}]$ for fine-tuning; batch size $[4, 32]$ for video; resolution $[224, 320]$; dropout $[0.2, 0.5]$; weight decay $[10^{-6}, 10^{-3}]$. For fusion: attention head sizes $[128, 512]$ and temperature scaling for calibration.

e) *Output (risk schema fields)*.: Emit `risk.deepfake` events extending Table III with:

- visual_score, audio_score, av_consistency_score
- media_hash (content hash), clip_seconds
- compression_level metadata (for robustness audit)

f) *Integration Role (feed correlation engine)*.: Deepfake evidence is correlated with identity anomalies and phishing/social engineering. Example: an incoming “CEO voice” call (media) flagged as suspicious plus concurrent anomalous authentication and an earlier phishing email increases incident severity and triggers a policy-based response (step-up authentication, block communication channels, or case escalation).

g) *Evaluation metrics and protocols*.: Primary: ROC-AUC, F1, balanced accuracy. Secondary: cross-dataset generalization gap, robustness under compression/noise, and calibration. For audio anti-spoof subtask, report metrics consistent with ASVspoof protocols where applicable [47].

h) *Five-week timeline (Mermaid Gantt)*.:



Fig. 5. System architecture and event flow.

i) *Expected contributions*.: (1) A multimodal deepfake detection plan emphasizing cross-dataset evaluation; (2) decomposed risk signals (audio/visual/consistency) to support correlation; (3) robustness reporting under compression and unseen manipulations.

j) *Literature review (10 sources; 1–2 sentence summary each)*.:

- 1) **FaceForensics++** [44]: Introduces a standardized benchmark across manipulation methods and compressions, establishing the need for robustness-focused evaluation.
- 2) **DFDC dataset** [45]: Provides a large-scale, diverse dataset and challenge framing for deepfake detection at scale; useful for generalization testing.
- 3) **FakeAVCeleb** [46]: Proposes an audio-video dataset with lip-synced fake audio and video; motivates multimodal fusion beyond visual-only detectors.
- 4) **ASVspoof 2019** [47]: Defines a benchmark for synthesized/converted/replayed speech and introduces evaluation considerations (including tandem cost metrics), supporting audio spoof modeling.
- 5) **LipForensics** [48]: Shows that semantic mouth-motion inconsistencies can improve generalization and robustness under distortions.
- 6) **MesoNet (WIFS 2018)** [50]: Provides a compact forgery detection network resilient to video compression constraints; serves as a lightweight baseline.
- 7) **FakeCatcher** [51]: Detects deepfakes via biological signals, illustrating alternative cues beyond artifacts and supporting robustness arguments.
- 8) **ART-AVDF (CVIU 2024)** [52]: Uses articulatory representation learning for audio-visual deepfake detection and reports improvements across DFDC/FakeAVCeleb-like benchmarks.
- 9) **ICCV 2025 AV synchronization** [53]: Proposes a two-stage multimodal framework addressing both intra-modal and cross-modal inconsistencies; motivates temporal consistency modeling.
- 10) **Survey of face manipulation/fake detection (Information Fusion 2020)** [54]: Provides a broad taxonomy of manipulation and detection approaches and highlights persistent challenges in generalization and evaluation.

7) *Member D Module (5-page contribution): Threat Correlation Engine & Autonomous Security Agent*:

a) *Module Title*.: **Centralized Threat Correlation Engine and Policy-Bound Autonomous Security Agent**.

b) *Problem Definition*.: Even strong per-domain detectors generate noisy, fragmented alerts. Security operations require correlation to group related events into incidents and reduce false positives, while response actions must be governed to prevent harmful over-automation. Threat modeling and analyst communication benefit from shared frameworks such as ATT&CK [8]. Incident response guidance emphasizes

systematic handling and continuous improvement [4], [5]. Empirical SOAR evaluation suggests automation can improve efficiency but may harm investigation completeness if misconfigured, motivating policy-bounded autonomy [55].

c) *Input Data (system interfaces).*: Member D consumes standardized risk events from Members A–C and optional context feeds:

- 1) `risk.nlp`: phishing/web-attack scores, extracted IoCs.
- 2) `risk.auth`: continuous authentication score, drift and takeover indicators.
- 3) `risk.deepfake`: audio/visual/consistency scores, media hashes.
- 4) Optional context (if available): asset criticality tiers, identity assurance levels, allowlists, and historical incident memory (subject to governance).

d) *Methodology (correlation, agent constraints, hyperparameters).*: **Correlation design (layered).**

- 1) **Normalization and deduplication**: enforce schema versioning and de-duplicate near-identical risk events.
- 2) **Fusion scoring**: compute incident confidence S using a calibrated fusion function (baseline: logistic regression or Bayesian update on calibrated module scores). Tunable weights $w_m \in [0, 3]$ and time-decay $\lambda \in [0.01, 0.2]$ for temporal relevance.
- 3) **Graph-based correlation**: represent events as a graph whose nodes are risk events and entities (subject, URL, domain, asset), with edges defined by time windows and shared indicators. Graph matching/contextualization approaches motivate higher-level grouping for SOC use [57].
- 4) **ATT&CK alignment**: map incident patterns to candidate tactics/techniques (e.g., Initial Access, Credential Access) for interpretability [9].

Autonomous response agent (policy-bound). Actions are selected from an allowlisted set with explicit approval thresholds:

- Low-risk actions (auto): enrich incident, create ticket, notify analyst.
- Medium-risk actions (conditional): block URL at proxy, quarantine email, enforce MFA step-up (with policy checks).
- High-risk actions (human approval): disable account, isolate host, revoke tokens.

Agent decisions are logged with policy IDs and evidence pointers; this follows incident handling guidance and emphasizes auditability [4]. For AI-specific threats to the system itself, ATLAS provides a structured knowledge base for adversarial AI tactics [10]. Model extraction threats motivate interface hardening and monitoring [16].

e) *Output (risk schema fields).*: Emit:

- `incidents`: `incident_id`, `incident_score`, `evidence_refs`, `ATT&CK candidates`, `recommended actions`.
- `actions`: `action_type`, `policy_id`, `approval_state`, `rollback_plan_ref`, `audit_ref`.

f) *Integration Role (system “glue”).*: Member D integrates all modules into an incident-centric surface. The correlation engine (i) reduces alert volume through grouping, (ii) improves confidence by combining weak signals, and (iii)

TABLE VIII
MEMBER D: METRICS PLAN (MODULE-LEVEL/SYSTEM-LEVEL).

Category	Metrics
Correlation utility	Alert reduction ratio; grouping accuracy
Operational latency	p50/p95 time-to-incident; time-to-action
Safety	Approval rates; false containment; audit completeness
Analyst quality	Ticket completeness proxy; error patterns

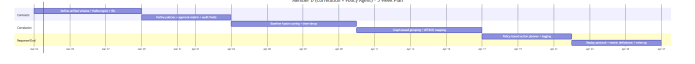


Fig. 6. System architecture and event flow.

generates consistent response recommendations with policy enforcement.

g) *Evaluation metrics and protocols.*: Because full system results are out-of-scope in this template-limited submission, we define *evaluation protocols*:

- 1) **Alert reduction ratio**: $\#risk\ events / \#incidents$.
- 2) **Incident grouping accuracy**: scenario-based evaluation using replayed event streams with known ground-truth grouping.
- 3) **Latency**: time from first risk event to incident creation; distribution p50/p95.
- 4) **Response safety**: rate of high-impact actions requiring approval; false containment rate under simulated benign scenarios.
- 5) **Analyst utility**: inspired by SOAR findings on efficiency vs. ticket completeness [55].

h) *Five-week timeline (Mermaid Gantt).*:

i) *Expected contributions.*: (1) A correlation and response design that is explicitly policy-bound and auditable; (2) incident-level metrics and replay protocol definitions; (3) ATT&CK-aligned incident representation enabling consistent analyst communication.

j) *Literature review (10 sources; 1–2 sentence summary each).*:

- 1) **MITRE ATT&CK** [8]: Provides a globally used taxonomy of adversary tactics/techniques grounded in real observations; supports interpretable mapping in correlation.
- 2) **ATT&CK tactics reference** [9]: Defines the “why” dimension of adversary behavior, supporting incident narrative and prioritization.
- 3) **NIST SP 800-61r3** [4]: Modern incident response guidance aligned with CSF 2.0 functions; motivates staged response and governance.
- 4) **NIST SP 800-137** [5]: Defines continuous monitoring strategy and risk visibility requirements; aligns with event-driven architectures.
- 5) **NIST AI RMF 1.0** [6]: Provides a lifecycle risk management framework for AI; motivates governance and evaluation beyond accuracy.
- 6) **MITRE ATLAS** [10]: Catalogs adversarial tactics against AI systems; guides threat modeling of AI-enabled SOC components.

- 7) **SOAR tools in use (C&S 2023)** [55]: Empirically evaluates SOAR tooling; shows efficiency gains and quality trade-offs, motivating bounded autonomy.
- 8) **SIEM kill-chain metadata aggregation (C&S 2020)** [56]: Proposes kill-chain-based alert metadata aggregation for SIEM improvement; motivates structured enrichment.
- 9) **Graph-based alert contextualization (2025)** [57]: Proposes representing alert groups as graphs and using graph matching for incident linking; motivates graph correlation design.
- 10) **DeepLog (CCS 2017)** [58]: Demonstrates sequence modeling for log anomaly detection and diagnosis, supporting the broader idea that temporal sequences enable stronger correlation than isolated alerts.

8) *Cross-module comparison tables (models, datasets, metrics)*:

VI. CONCLUSIONS, LIMITATIONS, FUTURE WORK, AND REFERENCES

A. Conclusions

This report defines a coherent research plan for an AI-driven multi-domain threat detection and response system by integrating four member-owned modules into a unified correlation-and-response pipeline. The core design choice is to emit standardized, calibrated risk events from heterogeneous detectors and fuse them into incident hypotheses that are interpretable (ATT&CK-aligned) and actionable under policy constraints. The plan explicitly addresses reproducibility and governance through a streaming microservice stack (Kafka/FastAPI/Docker) and MLOps controls (MLflow tracking and model registry), and it includes domain-appropriate evaluation protocols across text/web, behavioral authentication, audio-visual deepfakes, and SOC correlation.

B. Limitations

a) *Template-imposed limitations.*: This submission is intentionally restricted to the template’s Section 1 and Section 6, so it does not include full design/implementation chapters or empirical results chapters. Therefore, we specify evaluation *protocols* and *plans* rather than reporting executed experiments.

b) *Dataset realism and generalization.*: Public datasets can contain artifacts and distribution biases. For phishing and web attacks, email corpora and HTTP datasets may not represent current attacker language and infrastructure. For deepfakes, detectors can learn compression or dataset-specific cues rather than manipulation semantics; cross-dataset evaluation is required but may still not measure all in-the-wild manipulations. For behavioral biometrics, user drift and environmental variability may exceed benchmark settings, challenging real-world stability.

c) *Operational constraints and safety.*: Automation in response must be constrained. Empirical work shows automation can improve efficiency but also reduce investigation quality under certain conditions [55]. Inference services can become attack surfaces (e.g., model extraction) and require strict access control and output governance [16].

C. Future Work

Given a full capstone format beyond template constraints, future work should:

- 1) Implement the event-driven pipeline end-to-end (Kafka topics, microservices, correlation engine) and evaluate latency and throughput under replay.
- 2) Execute module training and systematic ablations (hyperparameters, features, calibrations, robustness).
- 3) Conduct cross-domain scenario evaluations (phishing → web exploitation → identity anomaly; deepfake impersonation → credential misuse) and measure incident grouping accuracy and false-positive reduction.
- 4) Formalize policy governance for the response agent using role-based approvals and integrate with a ticketing workflow; evaluate safety outcomes and analyst utility.
- 5) Extend the risk framework with Zero Trust-oriented context enrichment when organizational context is available [7].

D. Final responsibility summary (explicit)

Member A owns the `risk.nlp` module plan, including phishing/web datasets, NLP models, and evaluation protocol. **Member B** owns the `risk.auth` module plan, including behavioral biometrics datasets, verification metrics, and drift-aware evaluation.

Member C owns the `risk.deepfake` module plan, including AV datasets, multimodal architectures, and cross-dataset robustness evaluation.

Member D owns the correlation/agent plan, including unified schemas, incident mapping, policy constraints, and system-level correlation metrics.

E. References (IEEE style; direct URLs/DOIs included)

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TABLE IX
CROSS-MODULE SUMMARY: DATASETS, MODEL FAMILIES, AND PRIMARY METRICS.

Module	Domain	Primary datasets (examples)	Primary metrics
A	Email/HTTP	Enron, SpamAssassin, CIC-IDS2017, CSIC 2010	F1, ROC-AUC, ECE
B	Behavior	CMU Keystroke, Balabit Mouse	FAR, FRR, EER; delay
C	Media (AV)	FaceForensics++, DFDC, FakeAVCeleb, ASVspoof	ROC-AUC, F1; gen gap
D	SOC layer	Risk events from A/B/C + ATT&CK knowledge	Grouping acc.; latency; safety

TABLE X
CROSS-MODULE METRICS TABLE (AT-A-GLANCE).

Module	Metrics emphasis
A	Correctness + calibration on imbalanced security data
B	Verification-grade error rates + drift stability
C	Cross-dataset robustness + multimodal consistency
D	Operational utility + safety of automated actions

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